Module 16

Formal Relational Query Languages/1

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Week Recap

- SQL Examples have been practiced for basic query structures
- Nested Subquery in SQL
- Data Modification
- SQL expressions for Join and Views
- Transactions
- Integrity Constraints
- More data types in SQL
- Authorization in SQL
- Functions and Procedures in SQL
- Triggers

Module Objectives

- To understand formal query language through relational algebra

Module Outline

- Relational Algebra

Formal Relational Query Language

- Relational Algebra
  - Procedural and Algebra based
- Tuple Relational Calculus
  - Non-Procedural and Predicate Calculus based
- Domain Relational Calculus
  - Non-Procedural and Predicate Calculus based

Relational Algebra

Relational Algebra

- Created by Edgar F Codd at IBM in 1970
- Procedural language
- Six basic operators:
  - select: σ
  - project: Π
  - union: $\cup$
  - set difference: $-$
  - Cartesian product: $\times$
  - rename: ρ
- The operators take one or two relations as inputs and produce a new relation as a result

Select

Select Operation

- Notation: $\sigma_p(r)$
- p is called the selection predicate
- Defined as:

$$\sigma_p(r) = \{t \mid t \in r \text{ and } p(t)\}$$

  where p is a formula in propositional calculus consisting of terms connected by: $\wedge$ (and), $\vee$ (or), $\neg$ (not). Each term is one of:

$$\langle \text{attribute} \rangle \text{ op } \langle \text{attribute} \rangle \text{ or } \langle \text{constant} \rangle$$

  where op is one of: $=, \neq, >, \geq, <, \leq$
- Example of selection:

$$\sigma_{\text{dept_name}='\text{Physics}'}(\text{instructor})$$

Select Operation Examples

A	B	C	D
α	α	1	7
α	β	5	7
β	β	12	3

Project

Project Operation

- Notation: $\Pi_{A_1, A_2, \ldots, A_k}(r)$ where $A_1, A_2, \ldots, A_k$ are attribute names and r is a relation
- The result is defined as the relation of k columns obtained by erasing the columns that are not listed
- Duplicate rows removed from result, since relations are sets
- Example: To eliminate the `dept_name` attribute of `instructor`

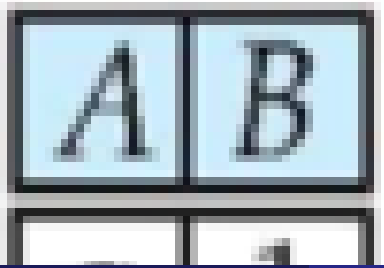
$$\Pi_{\text{ID, name, salary}}(\text{instructor})$$

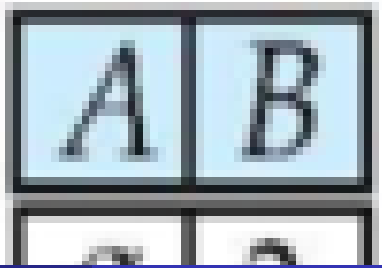
A	B	C
α	10	1
α	20	1

Union

Union Operation

- Notation: $r \cup s$
- Defined as: $r \cup s = \{t \mid t \in r \text{ or } t \in s\}$
- For $r \cup s$ to be valid:
  1. r, s must have the same arity (same number of attributes)
  2. The attribute domains must be compatible (example: 2nd column of r deals with the same type of values as does the 2nd column of s)
- Example: to find all courses taught in the Fall 2009 semester, or in the Spring 2010 semester, or in both





Difference

Difference Operation

- Notation: $r - s$
- Defined as: $r - s = \{t \mid t \in r \text{ and } t \notin s\}$
- Set differences must be taken between compatible relations
  - r and s must have the same arity
  - attribute domains of r and s must be compatible
- Example: to find all courses taught in the Fall 2009 semester, but not in the Spring 2010 semester

A	B
α	1

A	B
α	2

Intersection

Intersection Operation

- Notation: $r \cap s$
- Defined as: $r \cap s = \{t \mid t \in r \text{ and } t \in s\}$
- Assume: r, s have the same arity, attributes of r and s are compatible
- Note: $r \cap s = r - (r - s)$

A	B
α	1
α	2
β	1

A	B
α	2
β	3

s

A	B
α	2

Cartesian Product

Cartesian-Product Operation

- Notation: $r \times s$
- Defined as:

$$r \times s = \{tq \mid t \in r \text{ and } q \in s\}$$

- Assume that attributes of $r(R)$ and $s(S)$ are disjoint. (That is, $R \cap S = \emptyset$)
- If attributes of $r(R)$ and $s(S)$ are not disjoint, then renaming must be used

A	B
α	1
β	2

r

C	D	E
α	10	a
β	10	a
β	20	b

Rename

Rename Operation

- Allows us to name, and therefore to refer to, the results of relational-algebra expressions.
- Allows us to refer to a relation by more than one name.
- Example: $\rho_X(E)$ returns the expression E under the name X
- If a relational-algebra expression E has arity n, then

$$\rho_{X(A_1,A_2,\ldots,A_n)}(E)$$

returns the result of expression E under the name X, and with the attributes renamed to $A_1, A_2, \ldots, A_n$.

Division

Division Operation

- The division operation is applied to two relations
- $R(Z) \div S(X)$, where $X \subset Z$. Let $Y = Z - X$ (and hence $Z = X \cup Y$); that is, let Y be the set of attributes of R that are not attributes of S
- The result of DIVISION is a relation $T(Y)$ that includes a tuple t if tuples t_R appear in R with $t_R[Y] = t$, and with $t_R[X] = t_S$ for every tuple t_S in S.
- For a tuple t to appear in the result T of the DIVISION, the values in t must appear in R in combination with every tuple in S
- Division is a derived operation and can be expressed in terms of other operations:

$$r \div s \equiv \Pi_{R-S}(r) - \Pi_{R-S}((\Pi_{R-S}(r) \times s) - \Pi_{R-S,S}(r))$$

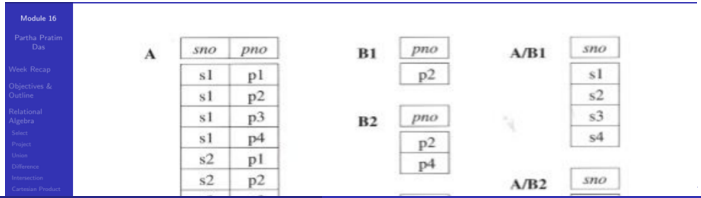
Division Examples

Subject
Prolog

Lecturer
Green
Lewis

Division Examples (2)

Lecturer
Green



Division Examples (3)

A	B

a	1
a	2
a	3

Division Example (4)

A	B	C	D	E
α	a	α	a	1
α	a	γ	a	1
α	a	γ	b	1
β	a	γ	a	1
β	a	γ	b	3
γ	a	γ	a	1
γ	a	γ	b	1
γ	a	β	b	1

r

- $r \div s$:

A	B	C
α	a	γ
γ	a	γ

- e.g. Students who have taken both 'a' and 'b' courses, with instructor '1'
- (Find students who have taken all courses given by instructor 1)

Module Summary

- Discussed relational algebra with examples

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